H
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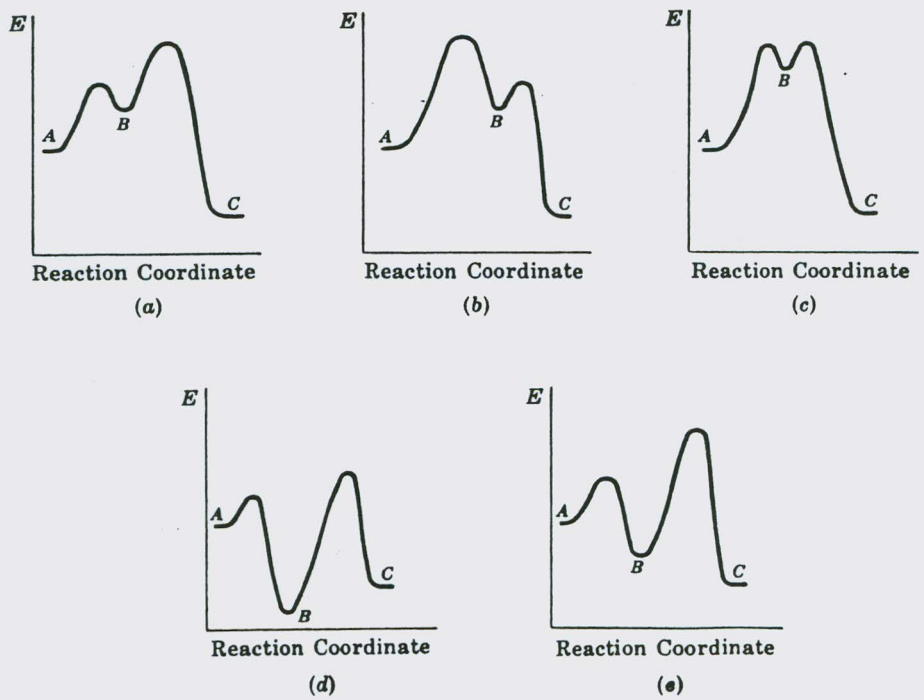
Physical Chemistry Cume
Kinetics

(1) A rule often quoted in the laboratory is "heat the sample by 10 °C and the reaction will proceed twice as fast." This is true at any given temperature. Under what condition is this true? Show all of your work.

(2) Using the reaction coordinate diagrams shown below, describe the relation of k_1 to k_2 (i.e., $k_1 \approx k_2$, k_1 less than k_2, or k_1 greater than k_2) for the consecutive reactions

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

where in each case the overall reaction $A \rightarrow C$ is exothermic.

Describe the possibility of isolation of B in each case as: little or fair or good or excellent.